MIL-D-2921B(SHIPS)
3 January 1964
SUPERSEDING
MIL-D-002921A(SHIPS)
8 February 1963
MIL-D-2921
24 August 1951
(See 6.3)

MILITARY SPECIFICATION

DISKS, RUBBER (CELLULAR, HARD)

1. SCOPE

1.1 Scope.- This specification covers hard, cellular, rubber disks.

1.2 Classification.- Rubber disks shall be of the following types, as specified (see 6.2):

Type I - Reclaimed rubber.
Type II - Synthetic rubber as the basic material.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-293 - Visual Inspection Guide for Cellular Rubber Items.

FEDERAL
FED-STD-601 - Rubber: Sampling and Testing.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N.Y.)

3. REQUIREMENTS

3.1 Material.- The material shall be as follows:

Type I - Utilizing 100 percent reclaimed rubber as the basic material.
Type II - Utilizing 100 percent synthetic rubber, either a copolymer product of butadiene and styrene or a copolymer of butadiene and acrylic nitrile.

3.2 Dimensions.- Unless otherwise specified, disks shall be 1.25-inches to 1.5-inches thick with a diameter not exceeding 6 inches, and shall have a concentric hole 5/8-inch to 3/4-inch in diameter through the center.

FSC 9320

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3.3 Density. - The density of a disk shall be not less than 8 nor more than 14 pounds per cubic foot.

3.4 Water absorption. - When tested as specified in 4.4.2, the increase in the weight of the disk shall not exceed the following:

Skin unbroken, 2 percent.
Skin removed from 2 parallel surfaces, 25 percent.

3.5 Structural strength. - When tested as specified in 4.4.3, disks shall withstand a compressive load of 150 pounds per square inch.

3.6 Impact strength. - The impact strength of the disks, when determined as specified in 4.4.4, shall be such that the depth of the impression left by the steel ball shall be not more than 0.150 inch.

3.7 Resistance to light-aging. - When subjected to the accelerated light-aging test specified in 4.4.5, disks shall not show injurious cracks, appreciable swelling, or other evidence of serious deterioration. When tested as specified in 4.4.4, the impressions left by the steel ball on the samples after the light-aging test shall be not more than 50 percent deeper than the impressions made before aging.

3.8 Heat resistance. - When tested as specified in 4.4.6, the density shall not increase more than 25 percent nor shall the volume increase more than 25 percent.

3.9 Workmanship. - The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection. - Sampling for quality conformance inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all material of the same type, submitted for delivery at one time.

4.2.1 Inspection of the end item. -

4.2.1.1 Examination of the end item. - Examination of the end item shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQLs) set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of disks for examination in 4.2.1.1.1, 4.2.1.1.2 and in units of shipping containers for examination in 4.2.1.1.3.

4.2.1.1.1 Examination of the end item for defects in appearance and workmanship. - The sample unit for this examination shall be one disk of each type.

Examine	Defect	Major	Minor
Appearance and Workmanship	Any classified defect as indicated in MIL-STD-293.		
	Presence of dirt, foreign material, imbedded particles.		x
	Distorted, not flat, faces not parallel.	x	
	Edges not smooth and perpendicular to face.		x

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4.2.1.1.2 Examination of the end item for dimensional defects. - The sample unit for this examination shall be one disk of each type.

Examine	Defect
Diameter	Over 6 inches.
Thickness	Less than 1.25 inch, more than 1.5 inches.
Hole	Less than 5/8 inch, more than 3/4 inch.
Concentricity of hole	Axes more than ±1/8 inch apart for three inch disk or more than ±1/4 inch apart for larger than six-inch disk.

4.2.1.1.3 Examination of preparation for delivery. - An examination shall be made to determine that packaging, packing and markings comply with the requirements of section 5 of this specification. The sample unit for this examination shall be one shipping container, fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

Examine	Defect
Packaging	Packaging material or methods not as specified.
Packing	Not level specified.
	Any nonconforming component, component missing, damaged or otherwise defective, affecting serviceability.
	Packing materials or methods not as specified.
Marking	Interior or exterior markings (as applicable) omitted, illegible, incorrect, incomplete or not in accordance with contract requirements.
Count	Less than specified or indicated number of disks.

4.2.2 Inspection levels and acceptable quality levels (AQLs) for examinations. - The inspection levels for determining the sample size and the acceptable quality levels (AQLs), expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection level	AQL	
		Major	Total
4.2.1.1.1	II	1.5	6.5
4.2.1.1.2	S-2	-	4.0
4.2.1.1.3	S-2	-	4.0

4.3 Testing of the end item. - The end item shall be tested for the applicable characteristics as indicated in table I on each lot presented for inspection. The sample unit shall be 11 disks. The inspection level for determination of sample size shall be S-1 except that no more than five sample units shall be randomly selected throughout the lot. There shall be no failure of the average or unit values to comply with the minimum requirement for each characteristic. The lot size shall be expressed in units of disks.

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Table I - Instructions for testing (sample unit).

Characteristic	Specification Reference		Rqmts. Appl. To	Number Determinations Per Sample Unit	Results Reported As	
	Requirement	Test Method	Indiv. Unit		Pass or Fail 1/	Numerically To Nearest 2/
Density	3.3	4.4.1	x	Avg of 3	-	0.1 lb percuft
Water absorption						
Skin unbroken	3.4	4.4.2	x	Avg of 3	-	0.1 percent
Skin broken	3.4	4.4.2	x	Avg of 3	-	1.0 percent
Structural strength	3.5	4.4.3	x	3	x	-
Impact strength	3.6	4.4.4	x	Avg of 3	-	0.005 inch
Light aging resistance						
Deterioration	3.7	4.4.5	x	3	x	-
Impact strength	3.7	4.4.5	x	Avg of 3	-	1 percent
Heat resistance						
Volume change	3.8	4.4.6	x	Avg of 3	-	0.1 percent
Density change	3.8	4.4.6	x	Avg of 3	-	0.1 lb percuft

1/ If failure is indicated report description of failure.

2/ Test reports shall include all values on which results are based.

4.4 Tests. -

4.4.1 Density. - The density of a disk shall be determined by dividing the weight of the disk in air by its volume. The volume shall be determined by measuring the volume of water displaced by the disk.

4.4.2 Water absorption test. -

4.4.2.1 Specimens. - Three test disks shall be used.

4.4.2.2 Procedure. - The weight of the disk in air shall be accurately determined. This weight shall be designated as W_1. The disk shall then be wetted with water and wiped. This weight to be designated as W_2. The disk shall then be subjected to immersion in water at 20°C. plus or minus 5°C. under a hydrostatic head of 8 pounds per square inch, plus or minus 0.5 pound for a period of 48 hours. The weight after immersion and wiping shall be designated as W_3.

$$\text{The percent water absorption} = \frac{W_3 - W_2}{W_1} \times 100.$$

4.4.2.3 Apparatus. - The vessel used to immerse the disk shall be so constructed as to facilitate the introduction and removal of sample. It shall be heavy enough to withstand 16 pounds per square inch pressure without appreciably changing in volume. The top of the vessel shall be so constructed that all air can be removed from the vessel. The top shall have a connection to a nitrogen or compressed air cylinder by way of a diaphragm valve. An accurate gage shall be installed in the tube line so that pressure may be maintained at a constant value even though the volume of water in the container is changing due to absorption by the disk. Three sets of determinations shall be made and the average reported. Considerable variation in water absorption shall be considered as indicating nonuniformity of the cellular structure and shall be reported.

4.4.3 Structural strength. -

4.4.3.1 Specimens. - Three test disks shall be used.

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4.4.3.2 Procedure. - A disk shall be placed between the platens of a standard compression testing machine and subjected to a gradually increasing load applied uniformly to the faces of the disk.

4.4.4 Impact strength. -

4.4.4.1 Specimens. - Three test disks shall be used.

4.4.4.2 Procedure. - The specimen shall be placed on a solid horizontal surface with one of the large faces upward. A steel ball 2 inches in diameter and weighing 1.179 pounds shall then be dropped on the surface from a height of 36 inches so as not to hit on the center hole. After the ball is dropped, the specimen shall be allowed to remain in a horizontal position for a period of 20 minutes before the effect of the impact is determined. At the end of the 20-minute period, the depth of the impression left by the steel ball shall be determined and recorded.

4.4.5 Accelerated light-aging tests. -

4.4.5.1 Specimens. - Three test disks shall be used.

4.4.5.2 Procedure. - The light-aging tests shall be conducted in accordance with method 7311 of FED-STD-601 with the following exceptions: There shall be no stretching, preconditioning or buffing of the specimens, and all eight filters shall be removed. (Total dosage in decomposition of oxalic acid, by uranyl oxalate actimometer, shall be equivalent to 4.0 by 10^4 mg. per square decimeter plus or minus 2 percent.)

4.4.6 Heat resistance test. -

4.4.6.1 Specimens. - The test specimens shall consist of 6 pieces 2 by 2 inches by thickness of disk.

4.4.6.1.1 Test specimens other than 2 by 2 inches by thickness of the disk may be cut from disks less than 6 inches outside diameter. The test specimens shall be a right angle parallelopiped having a minimum length or width of one inch.

4.4.6.2 Procedure. - The 6 specimens shall be suspended in a testing oven at 140°F. for 48 hours, making no immediate contact with metal parts. The apparatus shall be controlled so that the temperature will not vary more than plus or minus 2°F. The volume of all samples shall be determined before and after test by the water displacement method. The test specimens shall be weighed and the specific gravity computed before and after test.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early installation. -

5.1.1 Disks. -

5.1.1.1 Packaging. - Packaging shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name, and destination.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of packaging, packing and marking for shipment shall be specified (see 6.2).

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(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Packaging.-

5.2.1.1.1 Level A.- The disks shall be packaged in accordance with method III of MIL-P-116.

5.2.1.2 Packing (see 6.2).-

5.2.1.2.1 Level A.- Disk, packaged as specified (see 6.2), shall be packed in overseas type wood cleated fiberboard, nailed wood, wirebound wood, corrugated or solid fiberboard, wood cleated paper overlaid, or wood cleated plywood boxes conforming to PPP-B-591, PPP-B-621, PPP-B-585, PPP-B-636 (class 3), PPP-B-576 (class 2), or PPP-B-601, respectively, at the option of the contractor. Shipping containers shall have case liners conforming to MIL-L-10547 and shall be closed and sealed in accordance with the appendix thereto. Case liners for boxes conforming to PPP-B-636 may be omitted provided all joints and corners of the boxes are sealed with a minimum of 1-1/2 inches of pressure sensitive tape conforming to PPP-T-76. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds; that of fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.1.2.2 Level B.- Disk, packaged as specified (see 6.2), shall be packed in domestic type wood cleated fiberboard, nailed wood, wirebound wood, cleated plywood, or wood cleated paper overlaid boxes or class 2 fiber boxes conforming to PPP-B-591, PPP-B-621, PPP-B-585, PPP-B-601, PPP-B-576 (class 1), or PPP-B-636, respectively, at the option of the contractor. Box closures shall be as specified in the applicable box specification or appendix thereto. Thr gross weight of wood boxes shall not exceed 200 pounds; that of fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.1.3 Marking.- In addition to any special marking required by the contract or order or herein, interior packages and shipping containers shall be marked in accordance with MIL-STD-129 and shall include the date of manufacture (month and year).

6. NOTES

6.1 Intended use.- Disks covered by this specification are for use in the construction of buoyant items such as life nets and floats.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Packaging, packing or marking requirements other than those required by 5.1 (see 5.2).

6.3 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - Ships
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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION		
ORGANIZATION		CITY AND STATE
CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4063, WHICH IS OBSOLETE

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